

CARPORT A — 24' × 25' TWO-CAR (CLEAR SPAN)

Qty	Member	Section	Length
4	Girders (24' clear span @ 8'-4" o.c.)	3 × 12 × .188 6005A-T61 / 6061-T6	24'
3	Side beams (two 25' runs, splices on posts)	2 × 6 × 1/8 6061-T6	24'
4	Posts (8 pcs @ ~10', two per tube)	4 × 4 × 1/4 6061-T6	24'
9	Roof deck	1-1/2" Type B, 18 GA galvanized	27'

CARPORT B — 16' × 20' TRACTOR

Qty	Member	Section	Length
6	Rafters @ 4'-0" o.c.	2 × 8 × 1/8 6061-T6	18' (cut from 20')
2	Side beams (20' runs)	3 × 6 × 1/8 6061-T6	20'
3	Posts (6 pcs @ ~9', two per tube)	3 × 3 × 1/4 6061-T6	20'
6	Roof deck	1" Type B, 22 GA galvanized	22'

CONNECTIONS & HARDWARE (BOTH CARPORTS)

Qty	Item	Use
2 bars	Aluminum angle 3 × 3 × 1/4 6061-T6	Post saddle tabs + rafter clips (shop-cut)
1 bar	Aluminum flat bar 6" × 1/4 6061-T6	Post cap plates (saddles)
28	Bolts 1/2"-13 × 4" SS316 + nylock nuts + washers	Girder/beam to post saddles — 2 per post
1 box (250)	#14 × 1" SS self-drilling HWH	Rafter clips (4 per clip)
~750	#12-14 × 1-1/4" SS self-drilling + bonded EPDM washer	Deck to framing — every flute at perimeter/edges, per schedule in field
1 box	#10 × 3/4" SS stitch screws	Deck side laps @ 24" o.c. max
1 roll	Butyl tape 2"—3" (or bituminous paint)	Isolation: galvanized deck on aluminum flanges
56	Anchors Hilti Kwik HUS-EZ SS316 1/2" × 6"	Base plates — 4 per plate (screw-in, no expansion)

CONNECTION SYSTEM (AS DETAILED IN DRAWINGS)

Joint	Detail
Post → girder/beam	Shop-welded saddle (1/4" cap plate + two 1/4" × 4" tabs) — beam bears on post; (2) 1/2" SS316 through-bolts take wind uplift. Field work = set & bolt only.
Rafter → side beam	Aluminum angle clip 3×3×1/4 × 3"—4", (4) #14 SS screws per clip (or one 3/8" SS bolt per leg), uphill face.
Deck → framing	SS self-drillers with EPDM washers: every flute at perimeter, ends & corners ; field per schedule. Stitch screws at side laps. Butyl isolation tape on flanges.

Post → footing	Welded aluminum base plate + gussets, (4) Hilti KWIK HUS-EZ SS316 1/2×6 into 2'×2'×3' footing, top flush, smooth & level.
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*All shop welding in 6061/6005 aluminum; field connections mechanical (stainless) only. Sizes wind-checked for 170 mph / Exposure C open structure (ASD).
Final sealed engineering by Florida P.E. governs; this sheet and the attached drawings are the fabrication basis.*